

TECHIN

# JavaScript

## Ciklai

# JavaScript ciklai

```
1  // 1
2  while (condition) {
3    ...
4  }
5
6  // 2
7  do {
8    ...
9  } while (condition);
10
11 // 3
12 for(let i = 0; i < 10; i++) {
13   ...
14 }
```

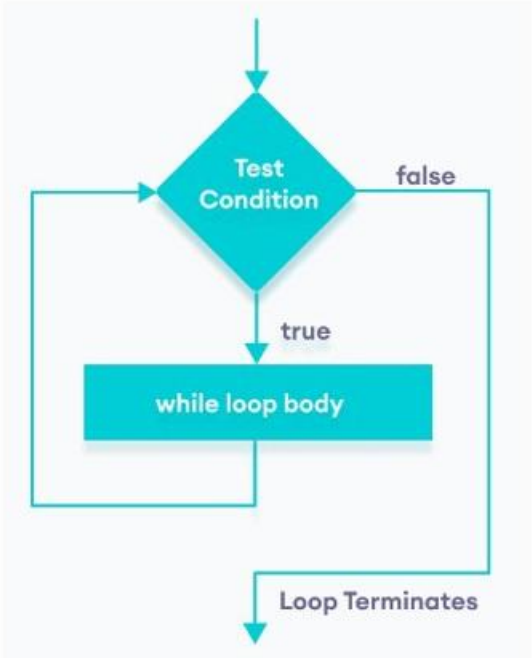
**Ciklas for darbui su objektais gali virsti į for...in**



# JavaScript ciklai

## while

```
while (condition) {
    // body of loop
}
```




```
// program to display numbers from 1 to 5
// initialize the variable
let i = 1, n = 5;

// while loop from i = 1 to 5
while (i <= n) {
    console.log(i);
    i += 1;
}
```

# JavaScript ciklai
## while

```
// program to find the sum of positive numbers
// if the user enters a negative numbers, the loop ends
// the negative number entered is not added to sum

let sum = 0;

// take input from the user
let number = parseInt(prompt('Enter a number: '));

while(number >= 0) {

    // add all positive numbers
    sum += number;

    // take input again if the number is positive
    number = parseInt(prompt('Enter a number: '));
}

// display the sum
console.log(`The sum is ${sum}.`);
```

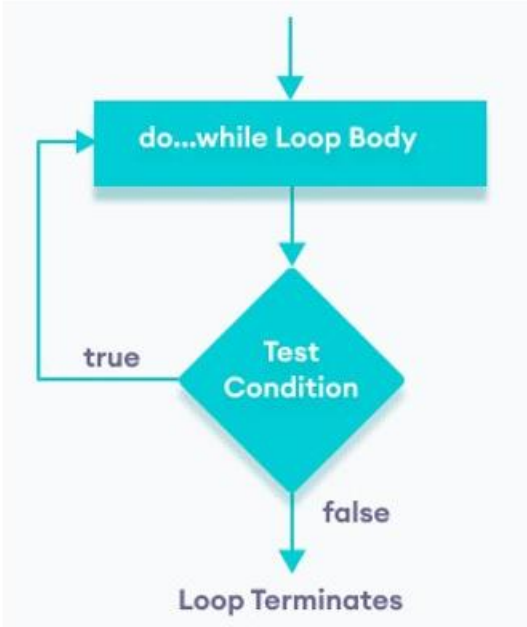
```
Enter a number: 2
Enter a number: 5
Enter a number: 7
Enter a number: 0
Enter a number: -3
The sum is 14.
```



# JavaScript ciklai

## do...while

```
do {
    // body of loop
} while(condition)
```




```
// to find the sum of positive numbers
// if the user enters negative number, the loop terminates
// negative number is not added to sum

let sum = 0;
let number = 0;

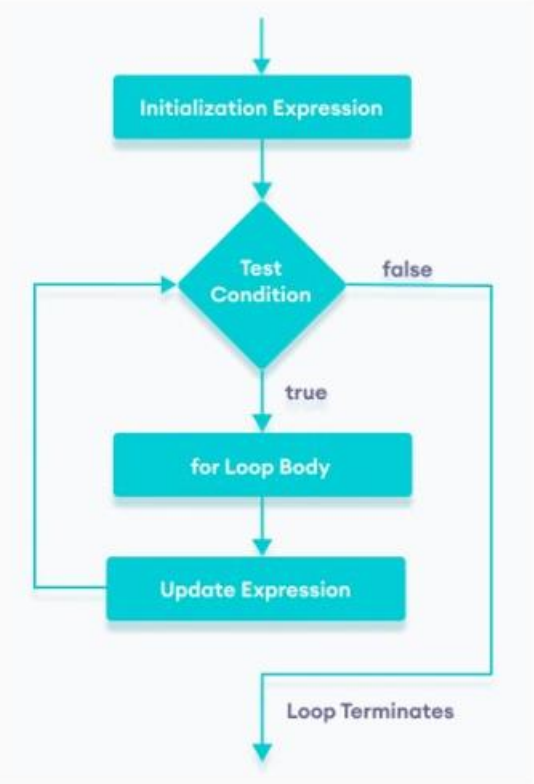
do {
    sum += number;
    number = parseInt(prompt('Enter a number: '));
} while(number >= 0)

console.log(`The sum is ${sum}.`);
```

```
Enter a number: 2
Enter a number: 4
Enter a number: -500
The sum is 6.
```

# JavaScript ciklai

## for




```
for (initialExpression; condition; updateExpression) {
    // for loop body
}
```

```
// program to display the sum of natural numbers
let sum = 0;
const n = 100

// looping from i = 1 to n
// in each iteration, i is increased by 1
for (let i = 1; i <= n; i++) {
    sum += i;  // sum = sum + i
}

console.log('sum:', sum);
```

```
sum: 5050
```

# Apibendrinkime...

- **while → naudokite tada, kai nežinote, kiek kartų reikės ciklui suktis (pvz., suksis tol, kol vartotojas įves 0)**
- **do...while → naudokite tada, kai kodą būtina paleisti bent vieną kartą, nesvarbu kas.**
- **for → naudokite visada, kai tiksliai žinote, kiek kartų ciklas turi suktis (pvz., 10 kartų arba per visą masyvo ilgį)**

# for...of

- **Skirtas masyvams.**
- **Einama per masyvą ir imami visi jo elementai vienas po kito.**

```
3   const vaisiai = ["Obuolys", "Bananas", "Kriaušė"];
4
5   for (const vaisius of vaisiai) {
6       console.log("Vaisius:", vaisius);
7   }
8
9   // Rezultatas: Obuolys, Bananas, Kriaušė
```

***FOR every fruit OF fruits***



# for...in

- **Skirtas objektams.**
- **Einama per objekto key, o tada galima lengvai gauti ir value.**

```
3   const auto = { marke: "Audi", spalva: "Juoda", metai: 2022 };
4
5   for (const savybe in auto) {
6       console.log(savybe + ": " + auto[savybe]);
7   }
8   /* Rezultatas:
9      marke: Audi
10     spalva: Juoda
11     metai: 2022
12  */
```



# JavaScript ciklai
## break;

- for ir while ciklą galima nutraukti su komanda break;

```
<script>
var text = "";
var i;
for (i = 0; i < 10; i++) {
  if (i === 3) { break; }
  text += "The number is " + i + "<br>";
}
document.getElementById("demo").innerHTML = text;
</script>
```

A loop with a **break** statement.

The number is 0
The number is 1
The number is 2



# JavaScript ciklai
## continue;

- **Vieną for ir while ciklo iteraciją galima praleisti su komanda continue;**

```
<script>
var text = "";
var i;
for (i = 0; i < 10; i++) {
  if (i === 3) { continue; }
  text += "The number is " + i + "<br>";
}
document.getElementById("demo").innerHTML = text;
</script>
```

A loop which will skip the step where i = 3.

The number is 0
The number is 1
The number is 2
The number is 4
The number is 5
The number is 6
The number is 7
The number is 8
The number is 9

